

DAVIS-MONTHAN AFB, AZ, USA

WMO: 722745

Lat: 32.167N

Lon: 110.883W

Elev: 824

StdP: 91.81

Time Zone: -7.00 (W07)

Period: 94-19

WBAN: 23109

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	0.1	1.7	-17.7	0.9	17.5	-15.2	1.1	18.4	10.6	16.2	9.1	14.4	2.5	150	0.463

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	40.7	18.5	39.5	18.4	38.2	18.2	22.7	29.6	22.2	29.9	21.8	30.1	5.0	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.3	17.6	24.5	20.4	16.6	24.8	19.7	15.9	25.3	71.1	29.2	69.1	29.6	67.4	29.8	26.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.3	8.2	7.3	-3.3	42.9	1.8	1.5	-4.6	44.0	-5.7	44.9	-6.7	45.8	-8.0	46.9		
			-5.4	24.1	2.0	1.0	-6.8	24.8	-8.0	25.4	-9.2	25.9	-10.6	26.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.2	11.9	13.1	16.5	19.6	24.3	29.8	30.7	30.1	27.8	22.1	16.4	11.5
	DBStd	7.78	3.70	3.77	3.73	4.00	3.53	2.69	2.32	2.11	2.35	3.77	4.08	3.72
	HDD10.0	73	23	13	3	1	0	0	0	0	0	0	4	28
	HDD18.3	769	199	148	78	33	3	0	0	0	0	13	82	213
	CDD10.0	4158	83	101	205	288	442	593	642	622	535	375	197	74
	CDD18.3	1812	1	2	22	70	187	343	384	364	285	130	24	1
	CDH23.3	25224	30	82	436	1088	2723	5394	5371	4814	3482	1467	317	19
	CDH26.7	14365	2	9	110	443	1493	3528	3286	2831	1926	658	79	1
Wind	WSAvg	3.4	3.1	3.2	3.4	3.7	3.6	3.6	3.5	3.3	3.3	3.3	3.3	3.1
Precipitation	PrecAvg	291	23	21	16	7	4	5	59	54	38	23	16	26
	PrecMax	555	122	83	56	34	28	40	173	125	147	126	56	128
	PrecMin	159	0	0	0	0	0	0	1	6	0	0	0	0
	PrecStd	80	23	21	14	8	7	8	39	31	32	30	14	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.2	28.0	31.5	35.0	39.0	42.5	42.4	41.2	38.9	35.9	31.2	25.3
		MCWB	10.8	11.2	12.6	14.1	15.8	18.1	19.1	20.3	18.8	16.1	13.4	10.7
	2%	DB	23.8	25.9	29.3	32.8	37.2	40.8	40.5	39.6	37.5	33.9	28.9	23.2
		MCWB	10.0	10.7	12.1	13.4	15.2	17.4	19.1	20.0	18.1	15.8	12.8	10.4
	5%	DB	21.9	23.9	27.7	31.2	35.3	39.7	39.1	38.0	36.3	32.4	27.1	21.3
		MCWB	9.5	10.3	11.5	12.8	14.6	17.2	19.4	19.9	18.3	15.4	12.4	9.9
	10%	DB	19.8	21.5	26.0	29.5	33.8	38.3	37.7	36.7	34.9	30.6	25.0	19.0
		MCWB	8.9	9.6	10.9	12.1	14.0	16.8	19.4	20.1	18.4	14.9	11.9	9.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.8	14.2	14.3	15.4	17.5	22.5	23.4	23.4	22.5	20.5	16.0	13.8
		MCDB	17.7	18.7	25.4	30.9	32.3	29.5	28.8	29.4	29.1	24.9	21.4	17.2
	2%	WB	12.1	12.8	13.1	14.3	16.3	20.8	22.6	22.8	21.8	18.9	14.7	12.2
		MCDB	18.3	19.6	24.7	29.9	32.5	31.3	29.3	29.8	28.5	26.9	23.7	19.4
	5%	WB	10.9	11.8	12.3	13.4	15.5	19.8	22.2	22.3	21.2	17.6	13.7	11.0
		MCDB	18.4	19.2	24.5	28.7	32.3	32.7	29.7	30.4	28.4	27.6	23.8	18.8
	10%	WB	9.9	10.7	11.5	12.6	14.7	18.7	21.7	21.8	20.6	16.5	12.7	10.0
		MCDB	17.9	19.4	24.0	27.5	31.8	34.4	29.9	30.7	28.5	27.1	23.1	17.6
Mean Daily Temperature Range	5% DB	MDBR	13.6	13.8	15.3	16.0	16.4	16.2	12.9	12.9	13.2	14.5	14.1	13.1
		MCDBR	16.4	17.3	17.8	18.1	18.0	17.1	15.0	14.7	14.8	16.4	16.3	15.5
		MCWBR	7.7	7.7	7.2	7.2	6.7	6.0	4.0	3.6	4.4	5.9	6.9	7.4
	5% WB	MCDBR	12.9	13.4	15.6	16.5	16.4	13.7	12.1	12.2	11.5	12.8	13.4	13.1
		MCWBR	6.5	6.4	6.4	6.7	6.0	4.6	2.9	3.2	3.4	4.9	6.1	6.7
Clear-Sky Solar Irradiance	taub	0.262	0.266	0.287	0.297	0.309	0.332	0.391	0.378	0.336	0.299	0.272	0.261	
	taud	2.634	2.628	2.536	2.495	2.499	2.439	2.406	2.431	2.519	2.567	2.601	2.628	
	Ebn at Noon	983	1007	998	993	976	947	891	901	933	952	960	966	
	Edh at Noon	75	83	99	108	108	115	118	114	100	88	77	72	
All-Sky Solar Radiation	RadAvg	3.60	4.54	6.03	7.41	8.20	8.20	6.81	6.45	5.92	5.05	3.98	3.29	
	RadStd	0.25	0.37	0.30	0.23	0.30	0.44	0.42	0.23	0.28	0.32	0.20	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (11 neighbors)	+0.27	N/A	N/A	N/A	N/A	N/A	N/A	-73	+106	N/A

Nomenclature: See separate page